

Correlating Social Media Metadata with Stock Volatility for Market Sentiment Analysis

AIT-582-DL1 | Dr. Can Nguyen
George Mason University
<https://mason.gmu.edu/~abagale/>

Abdul Basith
Volgenau School of Engineering
George Mason University
Fairfax, VA

abasith@gmu.edu

Akriti Bagale
Volgenau School of Engineering
George Mason University
Fairfax, VA

abagale@gmu.edu

Jason Chen
Volgenau School of Engineering
George Mason University
Fairfax, VA

Jchen77@gmu.edu

Qasim Ehsan
Volgenau School of Engineering
George Mason University
Fairfax, VA

qehsan@gmu.edu

Terrance Chiles
Volgenau School of Engineering
George Mason University
Fairfax, VA

tchiles@gmu.edu

Visvas Kaja
Volgenau School of Engineering
George Mason University
Fairfax, VA

vkaja@gmu.edu

Abstract— In this paper, we look at how user sentiment shared on social media can play a part in the volatility of the stock market. Social media has become a virtual town square where opinions can be shared to reach broad audiences. This in turn can mean trading on the stock market could be based on something trending on Reddit. We want to research this further and focus our data collection on two popular sites, X and Reddit, and using data from the year 2024. We ensure high data quality by focusing on posts deemed to be user generated, rather than spam or promotions (based on a predefined list of criteria we put together based on our observations). We use natural language preprocessing techniques such as tokenization and stopword removal to analyze the data. We presented our findings with visualization tools such as Matplotlib and NLTK for sentiment analysis.

Keywords—*social media, stock market, sentimental analysis models, natural language processing, X, Reddit*

I. INTRODUCTION

With the rise in use of social media, users generate vast quantities of data daily. Users frequently share their opinions about stocks on social media platforms such as X and Reddit. Analyzing these discussions can provide valuable insights into public sentiment regarding stocks. Additionally, there might be some correlation between market sentiment and the role it plays in influencing fluctuations in stock prices and overall market trends. These platforms enable users to share their views on trending topics and events that are currently happening at the moment. Due to the large number of users, the outreach can be substantial and can potentially influence outcomes. From the beginning, Reddit and X became the source of patterns, trends, and public sentiment. In the past, social media has influenced stock trades, politics, and other events.

In 2021, there was a large-scale event that took place in Reddit where investors in the subreddit r/WallStreetBets organized an extensive purchase campaign of GameStop stocks which made the price explode for a short period of time. This demonstrates the power of social media and its influence, as well as how these platforms can be used for change and manipulation. However, from this event occurring, there has been an underlying question that remains; does the sentiment of users from social media platforms truly affect stock market volatility? From the many users of platforms such as X or Reddit, there is most certainly some role that is being played to influence volatility of stocks, whether it be major or not.

II. BACKGROUND

In the age of the Internet, social media serves as an avenue for netizens to voice their opinions to wider audiences than previously. Before the internet, your primary community was limited to the surrounding several miles or as far as your publications could reach. The average person may not have been able to reach a large audience without money, fame, political influence, or familial influence. With the onset of the Internet, followed by social media websites such as Reddit, X, and Facebook, suddenly there was an opportunity for the public to share their views and reach wider audiences. This of course includes discussions about the market and the business decisions companies make. Having these public discussions available to anyone online means not only that their internal sentiments are now available to extract and analyze, but also that investors can be influenced to make decisions regarding their market behavior (i.e. investing into or selling stocks).

The specific social media platforms discussed in this paper will be Reddit and X (formerly known as Twitter). Reddit is a

popular platform where users share news articles, opinions, and analysis while driving discussion on various topics. Users can join “subreddits” regarding various topics. This platform has been known for sharing finance and investing advice, with subreddits like r/WallStreetBets gaining notoriety for driving action in the stock market. X is a blogging platform where users can post short, character-limited text posts called “tweets”. Other users can respond and requote tweets. X is a popular platform amongst all types of people, such as business owners, politicians, influencers, and the layman. On both these platforms users have the ability to share information in a way that spreads faster than traditional information sharing methods such as newspapers or journals.

III. PROBLEM DESCRIPTION

Traditional stock analysis methods often focus solely on numerical data, such as historical prices and trading volumes, while overlooking the impact of social media sentiment. Social media platforms, particularly X and Reddit, have become influential in shaping investor behavior and market movements. This project aims to bridge the gap between raw social media data and actionable financial insights by analyzing how social media sentiment correlates with stock volatility. For example, with the rise of retail investors and platforms like Reddit’s r/WallStreetBets, public sentiment can drive significant market events, as seen in the GameStop short squeeze of 2021.

This problem is important because public sentiment, understanding this relationship can provide valuable insights for investors, financial analysts, and policymakers. If there is an actual relationship between the public sentiment and stock prices, investors make informed investment decisions based on public perception rather than relying on the traditional financial metrics. Additionally, sentiment analysis can help financial institutions and regulators identify potential market volatility, allowing for better risk assessment and mitigation strategies. Studying social media sentiment helps us understand how psychological and emotional factors drive market movements, adding a new dimension to financial analysis. Due to the unpredictable nature of the stock market, it often requires lots of research to even get an idea of how the stock market will move. In the modern age where technology is prevalent, and the existence of social media allows users to obtain information extremely quickly as well as see the sentiment of others regarding the stock market.

Though the stock market cannot always be accurately predicted, the sentiment of the online masses may prove to be a useful tool in prediction as historically the market has shifted, evident by the 2021 GME stock surge. Major events such as elections, wars, etc. tend to affect both the stock market and public sentiment, this allows us to find correlations with how users of social media react and the volatility of the market. Though social media may not always be reliable, analyzing the overall sentiment in these platforms may bring about a wealth of information and can potentially be used to mitigate some uncertainty as well predict future movement

A. Hypothesis

Social media sentiment, particularly from platforms like Twitter and Reddit, has a significant correlation with stock price volatility. Positive or negative sentiment spikes, as well as high

engagement metrics, are likely to correspond with increased volatility in stock prices.

B. Research Questions

Research Questions:

1. What is the correlation between X/Reddit sentiment and stock market volatility?
2. Can sentiment analysis of social media posts predict short-term stock price fluctuations?
3. What role does the sentiment expressed in social media play in stock market anomalies?

IV. LITERATURE REVIEW

In recent years, the overlap between social media tweets/posts and the stock market has gotten more intertwined. This intersection between the two mediums has greatly reflected the emphasis the world of social media platforms has on the stock market. The stock market has been proven to be highly volatile because of several types of outcomes, but the social media landscape has been a major reason for the volatility in recent years. Although there are many platforms, social media websites like X, Reddit, and Discord have been able to access real time stock market prices and this allows the platforms to create many posts and discussion threads that ultimately leads to the change in the stock market.

Although social media sentiment can be a great driving force in the states of a company on the stock market, there are concerns about the misinformation from such platforms that can influence millions of users to make decisions that will eventually affect the stock market. These days, it is easy to create posts that claim to know something about the stock market even if the information is unverified. This review aims to look at how the social media age can affect the stock market all around the world.

Sentimental analysis aka "Opinion Mining" utilizes is a method to identify the author's feelings towards a topic. It uses Natural Language Processing (NLP) (Malawana & Rathnayaka, 2020). Social media text strings are fed into a sentimental analysis and scores are produced that fall under the categories or positive, neutral, and negative (Ivanenko, 2023). Sentimental analysis is interesting to apply to social media posts versus publications such as newspaper articles simply because social media posts are usually posted in the moment. While newspaper or other publications are thought out and curated carefully, social media posts give a direct look into what the public truly feels. Likewise, stock market fluctuations could also be due to the public's spontaneous decisions.

A survey by Reuters Institute in 2016 concluded that 51% of the respondents get all their news from various social media outlets (Jiao et al.). A lot of the news seen on social media can revolve around the constant state of the stock market. Whenever

social media makes big headlines for a certain company and their current position in the stock market, the market usually becomes volatile. But when a new media outlet reports similar findings, not much happens in the market, so it becomes clear that in recent years social media in particular is a significant driving force in stock market volatility.

Perhaps one of the biggest examples of this is the GameStop incident which occurred during the COVID-19 era. The economic state that the pandemic put the world through is certainly a factor, but the social media attention the company got is what made the stock market so volatile for the company. One of the conclusions made by Umar et al in their study about this event is how Reddit users caused a positive effect for GameStop since it brought a lot of attention from people around the country.

Another major example of public social media sentiment affecting the stock market volatility and prices would be the uncertainty of an ongoing or future election. As the political climate changes, so does future policies, usually those that are announced publicly by a political candidate through social media such as X (formerly known as Twitter). From historical data of past elections, in 2018 stocks went up substantially in certain fields while other sectors went down. The stock prices tend to drop before the election before going up after policies are made known, especially due to some presidents such as Donald Trump's vocal social media presence on industries they favor and not. According to a study conducted, US stock prices at the end of 2017 increased by a large margin (Blanchard et al., 2018). The methodology utilized was analysis of previous stock data before and after the election, however this growth can also be attributed to unexpected higher dividends. With the presence of social media especially X, it gives presidents or candidates power to announce their intentions with the market, thereby skewing public perception of certain industries which can also affect volatility.

V. DATA SOURCE:

This analysis draws upon two main data sources:

WallStreetBets 2022 Dataset (Kaggle)

The primary source of Reddit-based sentiment and discussion data is the WallStreetBets 2022 dataset, available on Kaggle. This dataset comprises user posts and comments from the popular r/WallStreetBets subreddit throughout the year 2024. It includes key features such as post titles, timestamps, upvotes, and comment counts, allowing for an in-depth exploration of retail investor sentiment and trending stock mentions. The dataset is particularly valuable for analyzing the influence of online communities on stock market movements and identifying retail trading trends.

Yahoo Finance (2025)

Stock market and financial data were sourced from Yahoo Finance, a widely used platform for real-time stock quotes, historical data, and financial news. This source provided accurate and up-to-date information on stock prices, historical trends, market capitalization, and other key financial indicators for the companies frequently mentioned in the Reddit dataset. Yahoo Finance served as the benchmark for aligning online sentiment with actual market performance.

VI. REASON FOR BIG DATA SOLUTION

Predictive models for market volatility can be improved with the use of metadata from social media. Although there is a significant level of noise in data from social media, utilizing ML tools to extract sentiment data from posts from Reddit and X can provide more timely market movement information. Analyzing publicly posted opinions about the market can be a more cost-effective method of making market predictions than relying on expert opinion or a more accurate prediction than relying on one's own analysis.

Since social media is easily accessible to virtually every country around the world, social media posts like tweets and posts can generate very large amounts of data. Data collected from a social media platform such as X or Reddit can be raw upon collection, so using the right tools to clean data and eventually perform a high-level analysis becomes important. Utilizing a big data framework allows us to analyze large data like these. To be able to efficiently extract any significant patterns and conclusions, Python based NLP libraries such as NLTK or spaCy become necessary. Furthermore, using sentimental analysis models such as VADER for text-based data becomes important because of the required computational load.

VI. PROPOSED APPROACH

As with any data driven research, the most important step we need to take is to collect valid data that can properly be used to give us statistically significant conclusions. Since the topic revolves around the idea of social media sentiment analysis, the type of data that we can use several tweets/posts from a major social media outlet such as X. Since we are correlating the sentiment analysis of social media with the stock market, we would also need to collect data regarding the stock market. The period of the posts collected needs to be consistent with the stock market as well, so we would have to make sure the months and years are the same for both data collected. For example, using the year 2020/2021 would be valid since a lot of tweets about the stock market took place during the pandemic, and stock market data from that year can show us a strong pattern as well. As with any stage in data cleansing, formatting the data and getting it ready for analysis is critical, so part of our approach is to clean the text from each tweet/post like removing the stop words and any special characters that would have no value to the overall meaning.

We could then use natural language processing methods to create a scoring system like Positive, Negative, and Neutral. Using tools like VADER or TextBlob makes this possible to do. Once this has been completed, we can then apply a machine learning model to analyze the social media sentiment. Analyzing data with all the proper tools and techniques with clean data has to be followed by validating the results. Testing the model's accuracy would be another step to confirm that our analysis is actually legitimate.

In terms of what tools to use, Python and R are usually the best languages for implementing such analysis so our approach would be to use a language such as Python which can be done across various platforms such as Jupyter notebook. Using Python gives us access to several libraries and packages that would make the analysis we aim to conduct possible. Using NLTK and SpaCy for text processing and other packages like TensorFlow or Keras for the modeling makes Python a strong approach for us.

VII. PROCESS OVERVIEW

A. Data Collection from Stock Market & Social Media

Our dataset comes from Kaggle and consists of r/WallStreetBets community post and comments. The dataset starts from 2022, and has been updated daily, allowing us to analyze discussions leading up to the present day if needed. We chose to focus specifically on 2024 for analysis. The dataset is in CSV format, allowing platform-independence, simplicity, and flexibility for data processing and visualization.

This dataset helps us to understand how retail investors discuss stocks, market trends, and economic conditions. It provides crucial elements that allow us to form relationships between online discussions and stock market behavior. Some of these elements include the title, score, number of comments, body, and timestamp of each post. The score is important by reflecting user engagement, helping us to determine which post accumulated the most interest and discussions. The timestamp allows us to map sentiment trends to specific dates in the market, enabling us to analyze how sentiment shifts align with stock price movements.

B. Pre-Processing and Data Cleaning

To ensure our analysis was accurate and meaningful. We implemented a comprehensive pre-processing and data cleaning workflow. Our goal was to refine the raw Reddit and stock market datasets to improve analytical quality and reduce noise. We focused on isolating relevant content from 2024, removing irrelevant or low-effort posts, and preparing the textual data for sentiment analysis. These steps helped merge the unstructured Reddit discussions with structured financial market data, enabling effective comparison and modeling.

Data Ingestion Approach:

Reddit data was ingested from a CSV file originally sourced from the WallStreetBets subreddit. This dataset included both structured metadata (e.g., post ID, timestamp, score, number of comments) and unstructured content (e.g., post titles and body text). The ingestion pipeline was designed to handle missing fields and redundant entries gracefully. Duplicate posts or threads were identified using post IDs and removed to prevent skewing the analysis with repeated content.

To assess the potential influence of Reddit sentiment on market performance, we also ingested historical stock market data for five companies: TSLA, AAPL, MSFT, AMC and GME. The yfinance Python library was used to retrieve publicly available pricing data from Yahoo Finance. Each ticker's data was stored in a CSV file containing the columns Open, High, Low, Close, and Volume. The daily market data enabled us to match Reddit sentiment with market movements at the same level of detail. The use of structured APIs ensured a reliable and automated ingestion process, minimizing manual handling or missing entries.

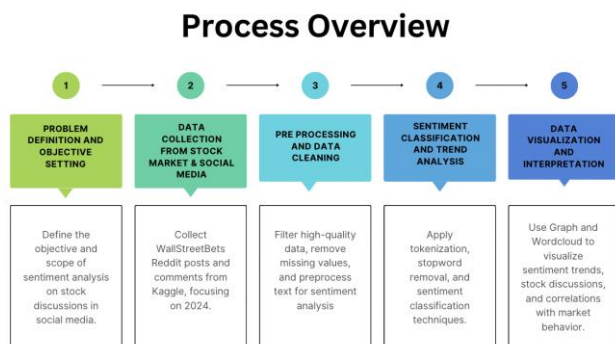


FIGURE 1: OVERVIEW OF THE PROCESS UTILIZED DURING THE ANALYSIS

Data Quality and Cleansing:

Ensuring high data quality was a top priority. We began by filtering out incomplete or low-effort content—specifically posts that lacked a title or a main body, which often provided insufficient context for reliable sentiment or thematic analysis. Posts with broken text encoding, empty comment threads, or spam-like content (e.g., posts with excessive links or promotions) were also discarded. This helped us maintain a corpus of meaningful user interactions.

In addition to structural filtering, score-based filtering was applied. We retained posts with a Reddit score greater than 50, based on the assumption that higher scores reflect more user engagement and community validation. This filtering step ensured that we focused on posts with broader visibility and impact, which are more likely to contain valuable discourse.

Furthermore, we prioritized posts that included both a title and comment section, as our exploratory analysis revealed that such posts often offered richer context and user interaction than standalone comments or questions.

Text Preprocessing:

For all retained textual data, we applied natural language preprocessing techniques including:

- **Tokenization:** Splitting sentences into individual words/tokens to facilitate analysis.
- **Stopword Removal:** Eliminating common but non-informative words (e.g., "the", "is", "and") to focus on more meaningful content.
- **Lowercasing:** Standardizing text for consistency during modeling.
- **Punctuation and Special Character Removal:** Cleaning out noise that could interfere with pattern recognition and sentiment classification.

These preprocessing steps helped improve the clarity and structure of the data, making it more amenable to downstream tasks such as topic modeling, sentiment analysis, and keyword extraction.

The stock market data retrieved via the yfinance API was also cleaned to ensure consistency. This included converting the 'Date' column into a proper datetime format, and removing currency symbols from numeric columns such as Open, High, Low, and Close so that they could be analyzed as floating-point values. These preprocessing steps allowed for accurate alignment with daily sentiment scores during the merging process.

VIII. MODEL DEVELOPMENT, TRAINING, AND TUNING

In Section III, we proposed a comprehensive modeling pipeline that begins by transforming the daily sentiment and price time series into lagged feature sets via a rolling-window

approach. We then perform time-series cross-validation to select the optimal lag order, using a k-fold split that respects temporal ordering and avoids look-ahead bias. Our regression suite includes both ordinary least squares (OLS) with an intercept term and a vector autoregression (VAR) framework, implemented using statsmodels and scikit-learn libraries. Hyperparameter tuning is achieved by grid-searching over lag values (1–5 days) and evaluating performance via mean absolute error (MAE) on held-out folds. Training is conducted on a standard desktop environment, with CPU-parallelized routines to reduce wall-clock time. We document runtime statistics, memory usage, and software versions, ensuring reproducibility. Model performance is quantified using RMSE, MAE, and directional accuracy metrics, and a simple “go-long” backtesting strategy is applied to assess economic significance and practical viability.

IX. RESULTS

Initial sentiment analysis showed increased positive sentiment from September to December (Figure 2), while negative sentiment remained consistent from January to September and began declining from October to December. For the stocks, overall sentiment was positive with Tesla being favorable, which is shown in Fig 3.

A. Data Visualization and Interpretation

To interpret sentiment trends over time, we employed several visualization techniques using Python’s Matplotlib library. We visualized the daily distribution of positive, negative, and neutral sentiment across Reddit posts to observe shifts in sentiment trends over time. We also examined sentiment distributions for specific stocks, such as TSLA, AAPL, MSFT, AMC, and GME. This will help to give insight on how often each company was associated with positive, negative, or neutral sentiments.

The preliminary results of sentiment analysis for 2024 (Fig 2) reveal a broad overview of public sentiment towards key topics and stock tickers, with significant fluctuations observed over the year. The sentiment seems to be mostly positive. The sentiment distribution for Reddit users, as shown in Fig 3, provides further insight into the public’s emotional tone, with positive, negative, and neutral sentiments mapped across various discussions. These scores highlight patterns in how Reddit users have responded to significant events or trends in 2024, offering a valuable perspective on social sentiment.

Fig 4 presents a bar chart of sentiment scores related to specific stock tickers, emphasizing how public sentiment correlates with market movements for various companies. This analysis provides a deeper understanding of how investor sentiment on platforms like Reddit can influence stock prices. In Fig 5, TSLA and AAPL showed noticeable upward trends, with TSLA experiencing significant growth in the final quarter of the year. MSFT maintained a relatively stable performance with modest fluctuations, while GME and AMC remained consistently low in price. This suggests that while some retail-favored stocks like

TSLA surged in popularity and valuation, others like AMC and GME did not reflect the same market momentum.

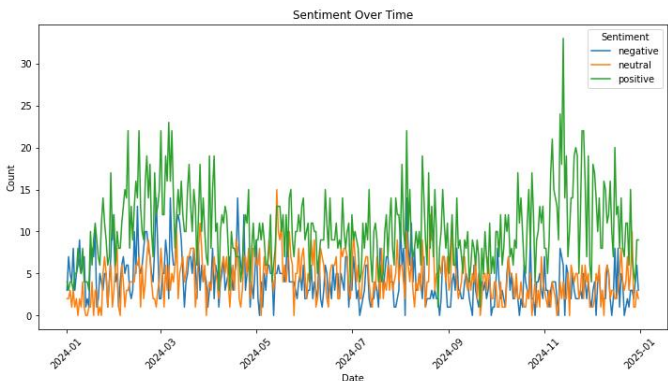


FIG 2: PRELIMINARY RESULTS OF SENTIMENT ANALYSIS FOR 2024

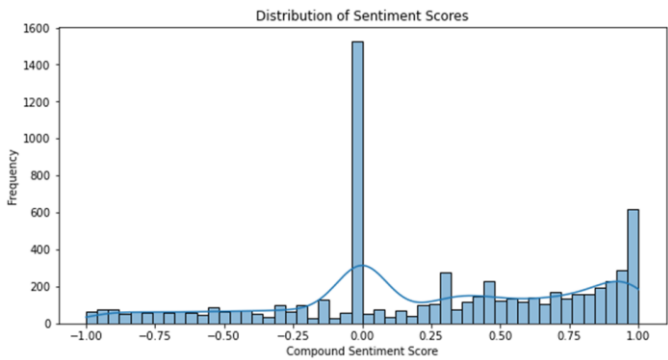


FIG 3: DISTRIBUTION OF REDDIT SENTIMENT SCORES (2024)

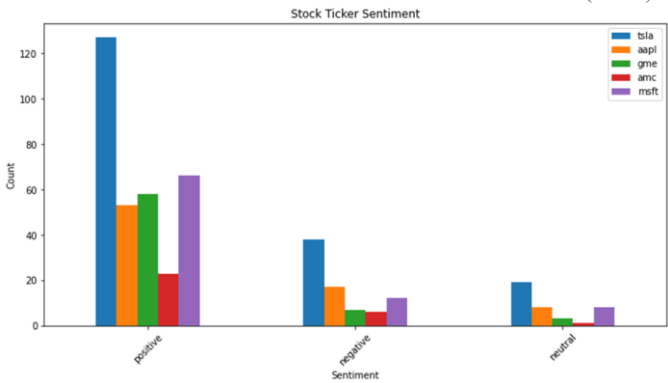


FIG 4: BAR CHAT OF STOCK TICKER SENTIMENT

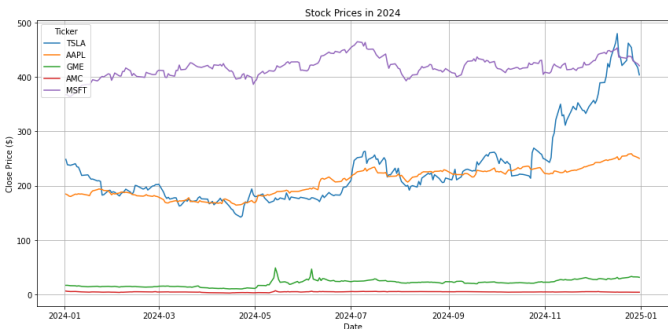


FIG 5: STOCK CLOSING PRICES (2024)

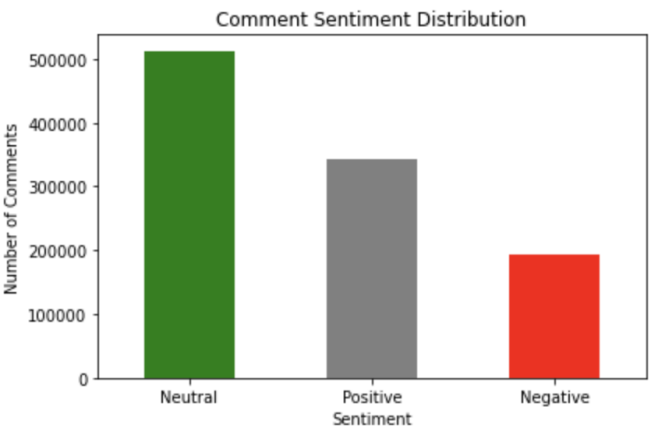


FIG 6: SENTIMENT OF REDDIT COMMENTS

Since the dataset from Reddit contains thousands of comments, it can be beneficial to have a visualization of how the comments show the overall sentiment. It can become difficult to find any sort of understanding when looking at thousands of comments without knowing what the nature of those comments are. So, libraries from Python such as TextBlob can be used to create a sentiment score for each comment which would be categorized into one of the three categories shown in Figure 6. Seeing the proportions of each sentiment category can help provide insights into the stock market performance.

Correlation Between Reddit Sentiment and Various Stock Price Trends (2024)

The following graphs compare Reddit sentiment trends with stock price movements for five popular stocks: AAPL, AMC, GME, MSFT, and TSLA. It highlights how online investor sentiment aligns — or diverges — from actual market behavior over time.

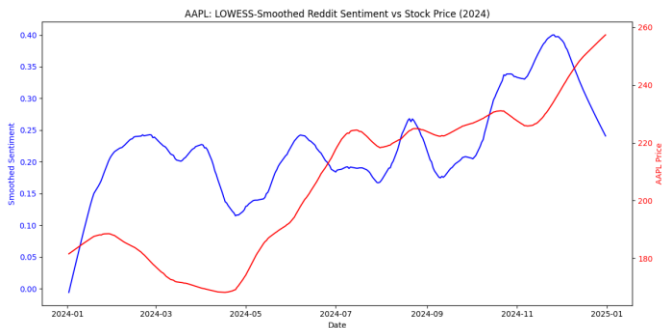


FIG 7: COMPARISON OF AAPL STOCK CLOSING PRICE AND REDDIT SENTIMENT SCORE OVER TIME (2024)

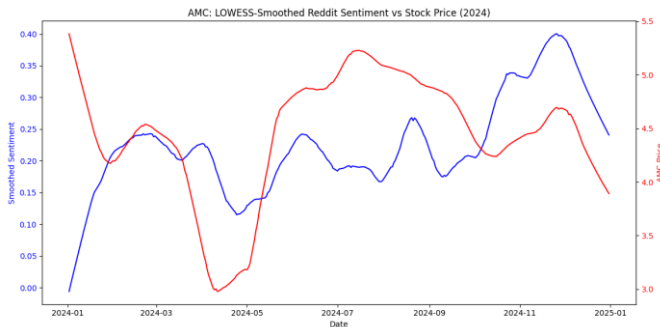


FIG 8: COMPARISON OF AMC STOCK CLOSING PRICE AND REDDIT SENTIMENT SCORE OVER TIME (2024)

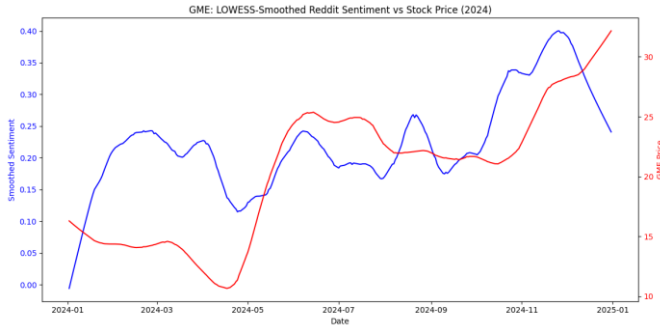


FIG 9: COMPARISON OF GME STOCK CLOSING PRICE AND REDDIT SENTIMENT SCORE OVER TIME (2024)

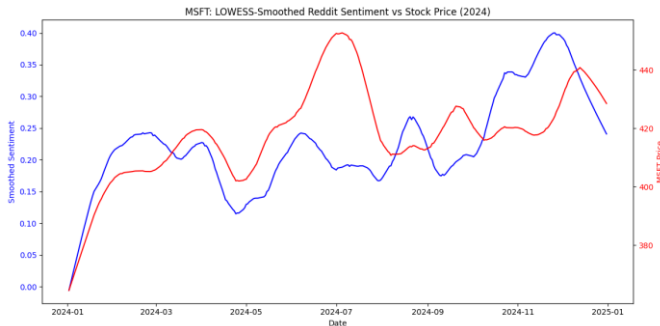


FIG 10: COMPARISON OF MSFT STOCK CLOSING PRICE AND REDDIT SENTIMENT SCORE OVER TIME (2024)

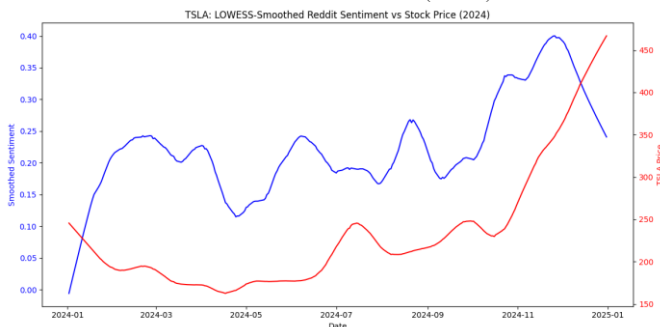


FIG 11: COMPARISON OF TSLA STOCK CLOSING PRICE AND REDDIT SENTIMENT SCORE OVER TIME (2024)

The analysis of Apple (AAPL) and AMC Theatres (AMC) highlights differing dynamics between Reddit sentiment and stock performance. AAPL shows a consistent upward trend in

both sentiment and stock price throughout the year, with sentiment slightly leading or moving in tandem with price increases—indicating strong alignment with overall investor optimism. On the other hand, AMC exhibits highly volatile sentiment with a flat or declining stock price, suggesting little to no correlation between Reddit discussions and market performance. This disconnect may reflect the fading influence of meme-driven hype on AMC's valuation.

The analysis of GameStop (GME), Microsoft (MSFT), and Tesla (TSLA) reveals distinct relationships between social media sentiment and stock price movements. For GME, sentiment spikes often coincide with sharp price changes, suggesting a partial explanation for its volatility driven by Reddit-driven, community-coordinated events. In contrast, MSFT shows a more stable pattern, where gradual sentiment increases align with a more steady price growth which indicates that Reddit sentiment may mirror broader investor confidence in the company. TSLA shows a case of high sentiment volatility that eventually aligns with strong stock performance in Q4, highlighting a moderate correlation where Reddit sentiment may both track and amplify market hype, especially for high-visibility stocks.

Some key takeaways from this is that sentiment significantly correlates with price movements in large-cap tech stocks like AAPL, TSLA, MSFT. Meme stocks (AMC, GME) show weaker or inconsistent relationships, suggesting external factors dominate. Our data supports the hypothesis that Reddit sentiment can reflect or influence stock behavior — especially for highly followed companies

Lagged Sentiment Regression

To explore the predictive relationship between Reddit sentiment and stock performance, we conducted a lagged regression analysis. The primary objective was to determine whether sentiment from the previous day could help forecast the next day's return for individual stocks. Daily returns were calculated using the formula $(Close_t \text{ minus } Close_{t-1}) \text{ divided by } Close_{t-1}$, and sentiment scores were shifted forward by one day to simulate their influence on subsequent price action. Sentiment values were also categorized into three groups: positive (greater than 0.5), neutral (between negative 0.5 and positive 0.5), and negative (less than negative 0.5) to support meaningful group-level comparisons. Each stock's analysis included two visual components. First, a boxplot illustrated how next-day returns varied across sentiment categories. Second, a scatterplot with a trendline plotted each day's sentiment score against the following day's return. The slope of the trendline indicated the direction of the relationship, while the correlation coefficient helped quantify its strength. Together, these visuals provided insight into whether sentiment offered any forward-looking signal for stock movement.

Tesla exhibited a trendline with a slight upward slope, suggesting that more positive sentiment may be associated with modest gains the next day. However, the relationship was weak. Microsoft showed a gentle downward slope, indicating that higher sentiment was sometimes followed by lower returns. This pattern hints at an inverse or contrarian effect. Apple's results showed a similar weak inverse relationship. The downward slope of the trendline suggests that higher Reddit sentiment is followed by slightly lower stock returns. However, the effect was subtle and does not indicate strong predictive value. For AMC and GameStop, the relationship between sentiment and return was erratic and lacked any consistent structure. This randomness reflects the speculative and volatile nature of these stocks, where external hype and social dynamics may overshadow measurable sentiment signals.

While the linear regression approach revealed weak relationships overall, the presence of minor patterns in Tesla and Microsoft suggests that sentiment might hold limited predictive value under certain conditions. More advanced methods such as time-aware machine learning models or further exploration of time lags could be used in future research to uncover more complex relationships between online sentiment and market behavior.

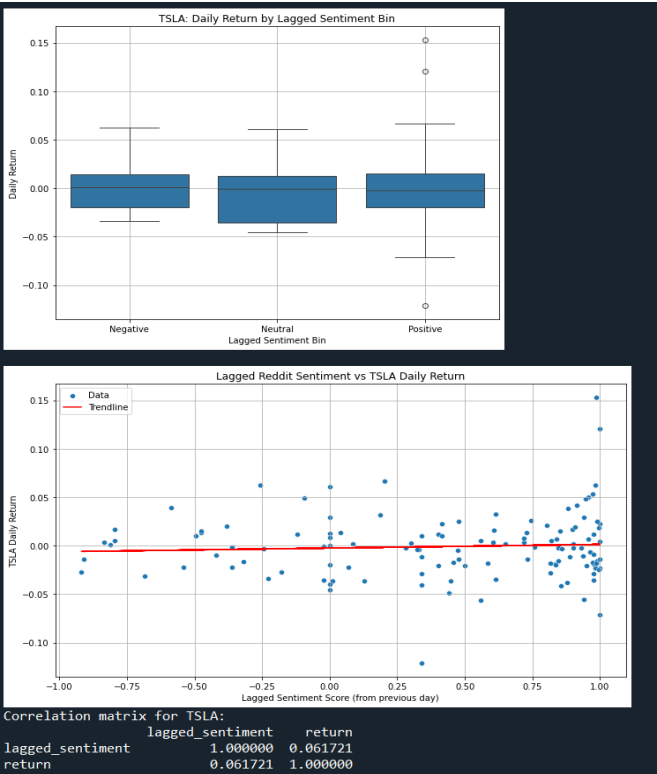


FIG 12: TESLA: DAILY RETURN VS LAGGED REDDIT SENTIMENT (WITH CORRELATION)

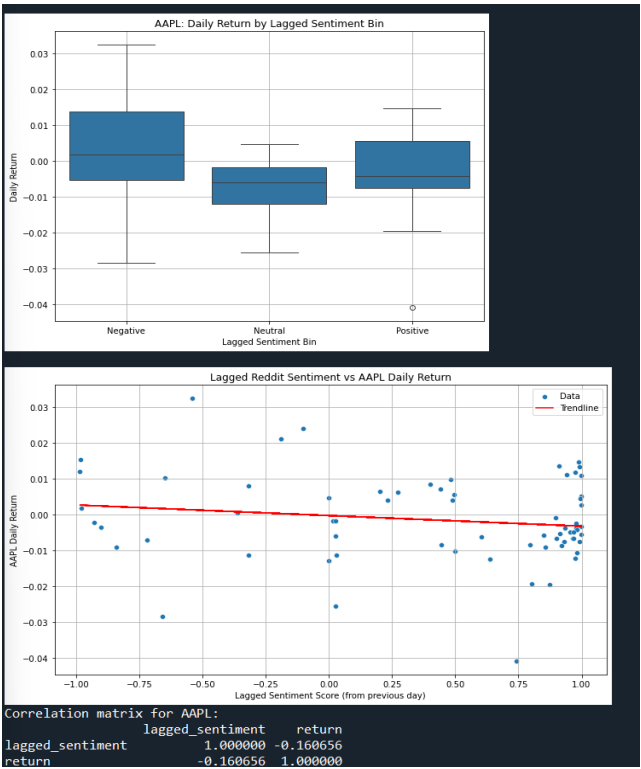


FIG 13: AAPL: DAILY RETURN VS LAGGED REDDIT SENTIMENT (WITH CORRELATION)

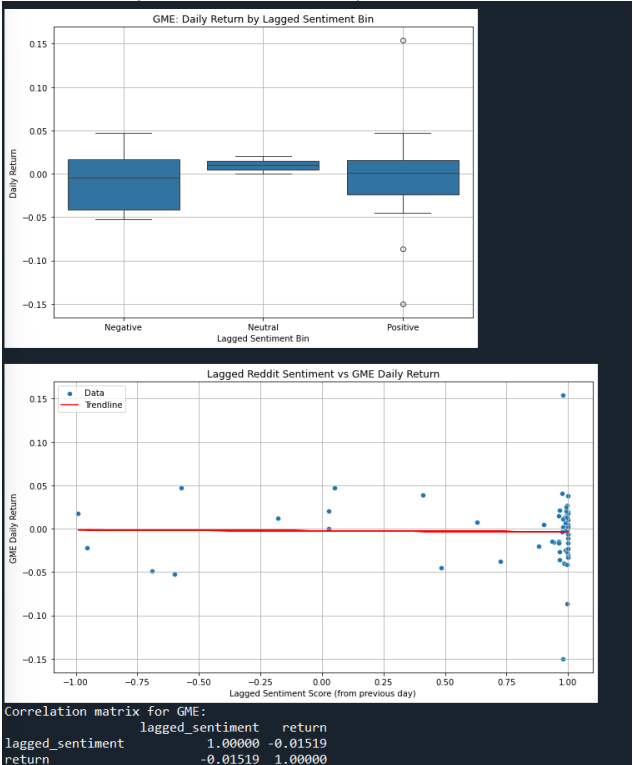


FIG 14: GME: DAILY RETURN VS LAGGED REDDIT SENTIMENT (WITH CORRELATION)

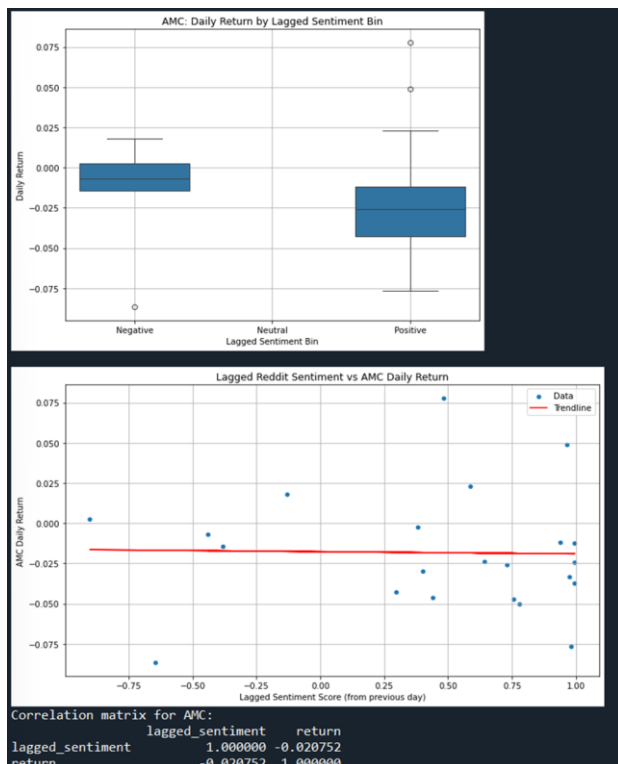


FIG 15: AMC: DAILY RETURN VS LAGGED REDDIT SENTIMENT (WITH CORRELATION)

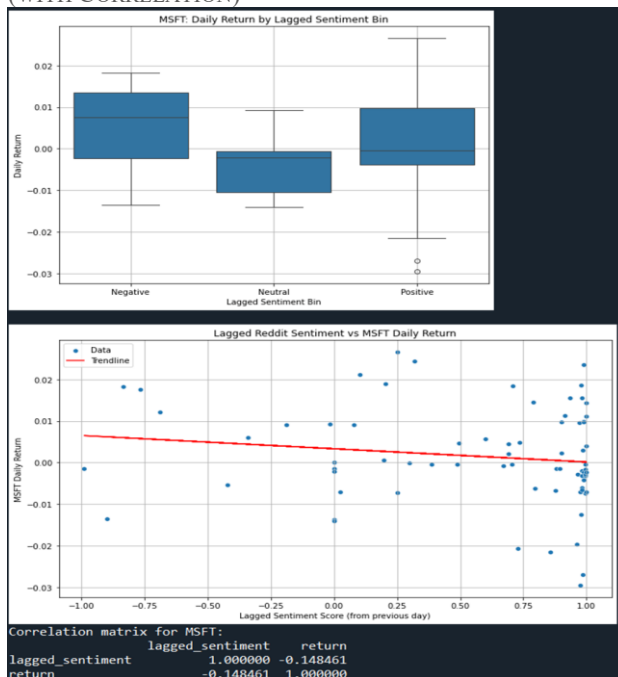


FIG 16: MSFT: DAILY RETURN VS LAGGED REDDIT SENTIMENT (WITH CORRELATION)

X. FUTURE WORK

Future research could benefit from expanding beyond Reddit to include additional sources of social media sentiment. The strong stock discussion focus and high retail trader engagement on WallStreetBets made it an excellent starting point but Twitter data integration would enhance public

sentiment analysis by providing more diverse perspectives. While we attempted to include Twitter data, API restrictions limited both the number of retrievable tweets and the historical range, presenting a challenge that future work should aim to overcome. Another important direction is experimenting with different timeframes measuring the relationship between sentiment and market performance. Our current analysis focused on next-day returns, but future studies could explore whether sentiment aligns more closely with same-day activity, weekly performance, or even longer-term trends. Adjusting the time horizon may reveal stronger predictive relationships.

In addition, future studies could introduce more advanced modeling techniques. Machine learning methods that account for nonlinear relationships and include additional financial indicators such as trading volume or volatility may improve prediction accuracy and uncover deeper patterns between online sentiment and stock market behavior.

REFERENCES

- Blanchard, O., Collins, C. G., Jahan-Parvar, M. R., Pellet, T., & Wilson, B. A. (2018). A year of rising dangerously? The U.S. stock market performance in the aftermath of the presidential election. *Journal of Policy Modeling*, 40(3), 489–502. <https://doi.org/10.1016/j.jpolmod.2018.03.004>
- Ivanenko, V. (2023). Assessing social media's impact on stock market predictions: Financial sentiment embedding approaches. 2023 13th International Conference on Dependable Systems, Services and Technologies (DESSERT), 1–6. <https://doi.org/10.1109/DESSERT61349.2023.10416463>
- Malawana, M. V. D. H. P., & Rathnayaka, R. M. K. T. (2020). The public sentiment analysis within big data distributed system for stock market prediction: A case study on Colombo Stock Exchange. 2020 5th International Conference on Information Technology Research (ICITR), 1–6. <https://doi.org/10.1109/ICITR51448.2020.9310871>
- Jiao, P., Veiga, A., & Walther, A. (2020). Social media, news media and the stock market. *Journal of Economic Behavior & Organization*, 176, 63–90. <https://doi.org/10.1016/j.jebo.2020.03.002>
- Umar, Z., Gubareva, M., Yousaf, I., & Ali, S. (2021). A tale of company fundamentals vs. sentiment-driven pricing: The case of GameStop. *Journal of Behavioral and Experimental Finance*, 30, 100501. <https://doi.org/10.1016/j.jbef.2021.100501>

LIBRARIES USED IN THE CODE

C. R. Harris et al., “Array programming with NumPy,” *Nature*, vol. 585, no. 7825, pp. 357–362, 2020. [Online]. Available: <https://doi.org/10.1038/s41586-020-2649-2>

W. McKinney, “Data structures for statistical computing in Python,” in *Proc. 9th Python in Science Conf.*, 2010, pp. 56–61. [Online]. Available: <https://doi.org/10.25080/Majora-92bfl922-00a>

J. D. Hunter, “Matplotlib: A 2D graphics environment,” Comput. Sci. Eng., vol. 9, no. 3, pp. 90–95, 2007. [Online]. Available: <https://doi.org/10.1109/MCSE.2007.55>

M. L. Waskom, “Seaborn: Statistical data visualization,” J. Open Source Softw., vol. 6, no. 60, p. 3021, 2021. [Online]. Available: <https://doi.org/10.21105/joss.03021>

P. Virtanen et al., “SciPy 1.0: Fundamental algorithms for scientific computing in Python,” Nat. Methods, vol. 17, pp. 261–272, 2020. [Online]. Available: <https://doi.org/10.1038/s41592-019-0686-2>

S. Bird, E. Klein, and E. Loper, *Natural Language Processing with Python*. Sebastopol, CA, USA: O’Reilly Media, 2009. [Online]. Available: <https://www.nltk.org>

Python Software Foundation, “The Python Standard Library: os — Miscellaneous operating system interfaces.” [Online]. Available: <https://docs.python.org/3/library/os.html>

APPENDIX

INTERPRETATION OF REDDIT SENTIMENT VS. STOCK PRICES (2024)

OVERVIEW OF CORRELATION RESULTS

Ticker	Pearson Correlation	P-value	Interpretation
AAPL	0.3355	0.0000	Moderate positive correlation — statistically significant
AMC	0.0334	0.5979	Near zero correlation — not significant
GME	0.2494	0.0001	Weak to moderate correlation — significant
MSFT	0.2629	0.0000	Weak to moderate correlation — significant
TSLA	0.3324	0.0000	Moderate correlation — significant

Note: A Pearson correlation ranges from -1 (perfect negative) to +1 (perfect positive). A value near 0 means no correlation.